SQL Fundamentals for Statistical Analysis

Training handout — sample material prepared for this prototype.

1. Relational Data in Statistical Work

A relational database stores data as tables. Each table holds rows, and each row holds one observation. In survey work a row is typically one household, one member of a household, or one enterprise. Each column holds one variable, such as district code, monthly consumption expenditure, or principal activity status.

Every table should have a primary key: a column, or a combination of columns, whose value is unique for every row and never null. In a household schedule the primary key is usually the combination of the first-stage unit identifier, the household serial number and the survey round. A foreign key is a column in one table whose values refer to the primary key of another table. Foreign keys are what make joins meaningful, because they record which rows belong together.

Structured Query Language, SQL, is the language used to ask questions of a relational database. A statistical officer who can write SQL can produce a tabulation directly from the survey database instead of waiting for a programmer to prepare an extract. That independence is the practical reason SQL sits in the competency framework.

2. SELECT: Choosing Columns and Rows

The SELECT statement retrieves data. Its simplest form names the columns wanted and the table they come from. Writing SELECT followed by an asterisk returns every column, which is convenient when exploring a table but wasteful in production queries, because the database must read and transmit columns nobody will use.

The FROM clause names the source table. The optional WHERE clause restricts which rows are returned. A query without a WHERE clause returns every row in the table, which for a national survey may be several hundred thousand records.

Column aliases, introduced with the keyword AS, rename a column in the output. This matters when a query result is going directly into a published table, because the alias becomes the column heading. SELECT DISTINCT removes duplicate rows from the result, which is the usual way to list the set of district codes actually present in a sample.

3. Filtering with WHERE

The WHERE clause takes a condition and keeps only the rows for which the condition is true. Comparison operators are equals, not equals, greater than, less than, greater than or equal to, and less than or equal to. Conditions combine with AND, OR and NOT.

The BETWEEN operator tests whether a value falls within an inclusive range, which is the natural way to filter an age group or an expenditure class. The IN operator tests membership of a list, and is the readable way to restrict a query to a set of state codes. The LIKE operator performs pattern matching on text, using the percent sign to stand for any sequence of characters and the underscore to stand for exactly one character.

Missing data needs care. In SQL a null value means unknown, not zero and not an empty string. Comparing anything to null with the equals operator yields unknown rather than true, so a row with a null value is not returned even by a condition that appears to test for it. To test for missing data you must write IS NULL or IS NOT NULL. This is the single most common source of silently wrong tabulations in survey processing, because non-response is recorded as null.

4. Aggregate Functions

Aggregate functions collapse many rows into a single value. COUNT returns the number of rows. SUM adds a numeric column. AVG returns the arithmetic mean. MIN and MAX return the smallest and largest values present.

The distinction between COUNT with an asterisk and COUNT applied to a named column is important and frequently missed. COUNT with an asterisk counts rows, including rows where every value is null. COUNT applied to a named column counts only the rows where that column is not null. Reporting the second when the first was intended understates a denominator, and understating a denominator inflates every rate computed from it.

AVG likewise ignores null values entirely rather than treating them as zero. If a consumption variable is null for households that did not respond, AVG returns the mean of the responding households, which is not the same quantity as the mean over all sampled households. Deciding which of the two is wanted is a methodological judgement, not a SQL question, but the officer writing the query has to know that the choice is being made.

5. GROUP BY and HAVING

GROUP BY divides rows into groups that share the same value in the named columns, and applies aggregate functions within each group separately. Grouping a household table by district code and applying AVG to monthly expenditure produces one mean per district. Any column in the SELECT list that is not inside an aggregate function must appear in the GROUP BY clause; otherwise the database cannot know which of the many values in the group to display.

HAVING filters groups after aggregation. WHERE filters rows before aggregation. This is the difference that matters most, and it is the one most often confused. A condition on an individual household, such as excluding households with no members, belongs in WHERE. A condition on a computed group value, such as keeping only districts where at least thirty households responded, belongs in HAVING, because the count does not exist until aggregation has happened.

The two clauses can appear in the same query and often should. A typical tabulation filters out ineligible records with WHERE, groups the remainder by district, and then suppresses small cells with HAVING. The order in which the database evaluates the clauses is FROM, then WHERE, then GROUP BY, then HAVING, then SELECT, and finally ORDER BY.

6. Join Types

A join combines rows from two tables using a condition, almost always an equality between a foreign key and a primary key. The join type determines what happens to rows that have no match on the other side, and choosing the wrong type is how records disappear from a tabulation without anyone noticing.

An INNER JOIN returns only the rows that have a match in both tables. Joining a household table to a member table with an INNER JOIN drops any household that recorded no members. That may be correct, but it is a decision, and it silently changes the denominator.

A LEFT JOIN, sometimes written LEFT OUTER JOIN, returns every row from the left table, together with matching rows from the right table where they exist. Where no match exists, the columns from the right table are filled with nulls. This is the join to use when the left table is the population of interest and the right table supplies optional detail, because it guarantees the row count of the left table is preserved.

A RIGHT JOIN is the mirror image, keeping every row from the right table. A FULL OUTER JOIN keeps unmatched rows from both sides, filling the absent side with nulls, and is the natural tool for reconciling two registers that are each supposed to be complete. A CROSS JOIN returns every combination of rows from both tables and is almost never what a statistical query wants; encountering an unexpectedly enormous result usually means a join condition was omitted.

7. Ordering, Limiting and Reading a Query

ORDER BY sorts the result. The default direction is ascending; DESC reverses it. Sorting happens after aggregation, so a query can order districts by their computed mean expenditure. LIMIT restricts the number of rows returned, which is useful for inspecting a large result before running the full tabulation.

When reading an unfamiliar query, work outward from the FROM clause rather than reading top to bottom. Establish which tables are involved and how they are joined, then find the WHERE conditions that reduce the rows, then the grouping, then the aggregates. The SELECT list, although written first, is evaluated almost last and is usually the least informative part of the query.

Finally, a query that runs is not the same as a query that is correct. Before publishing any number produced by SQL, check the row count against an independent source, confirm that nulls have been handled deliberately, and verify that the join type preserved the intended denominator.